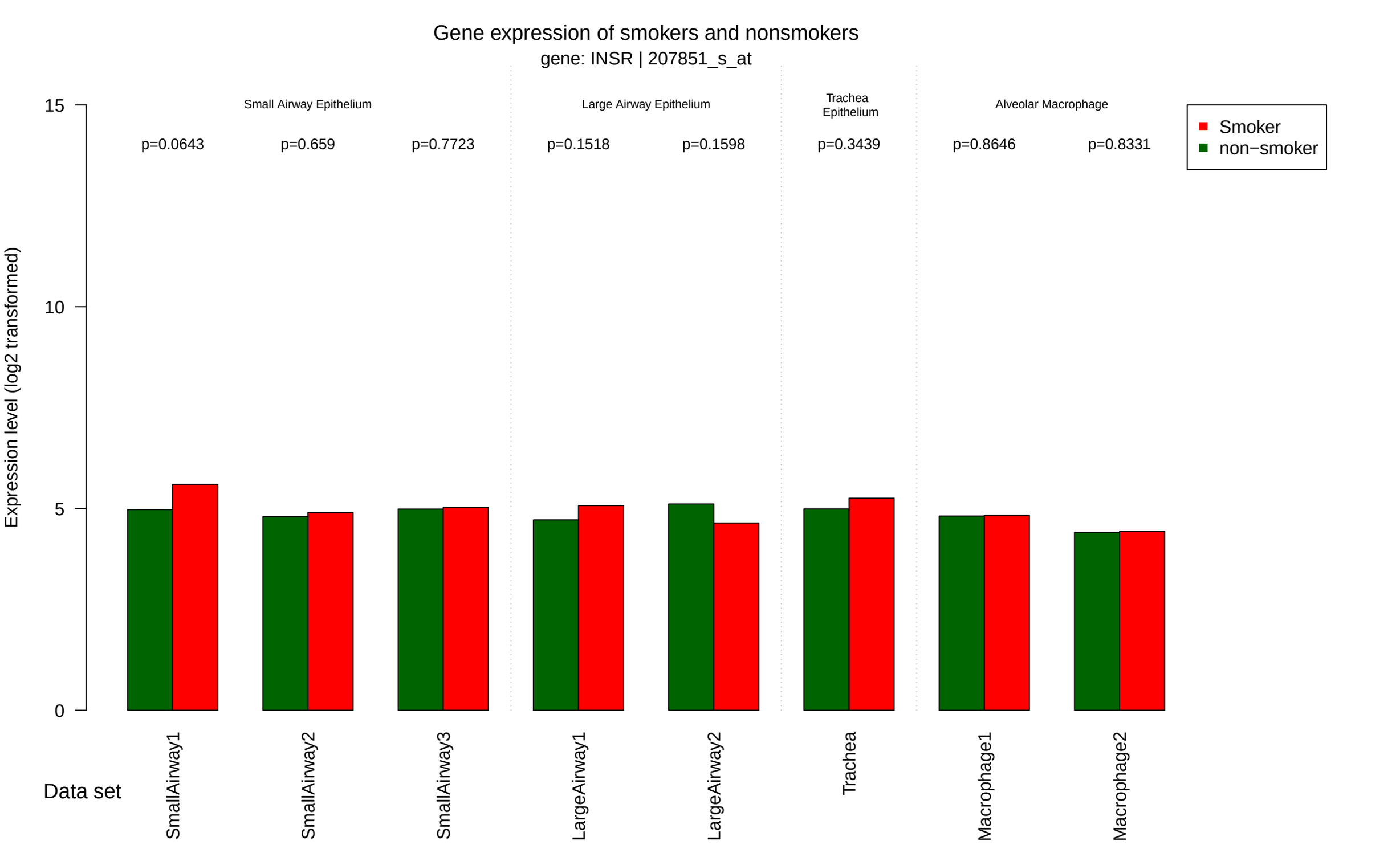
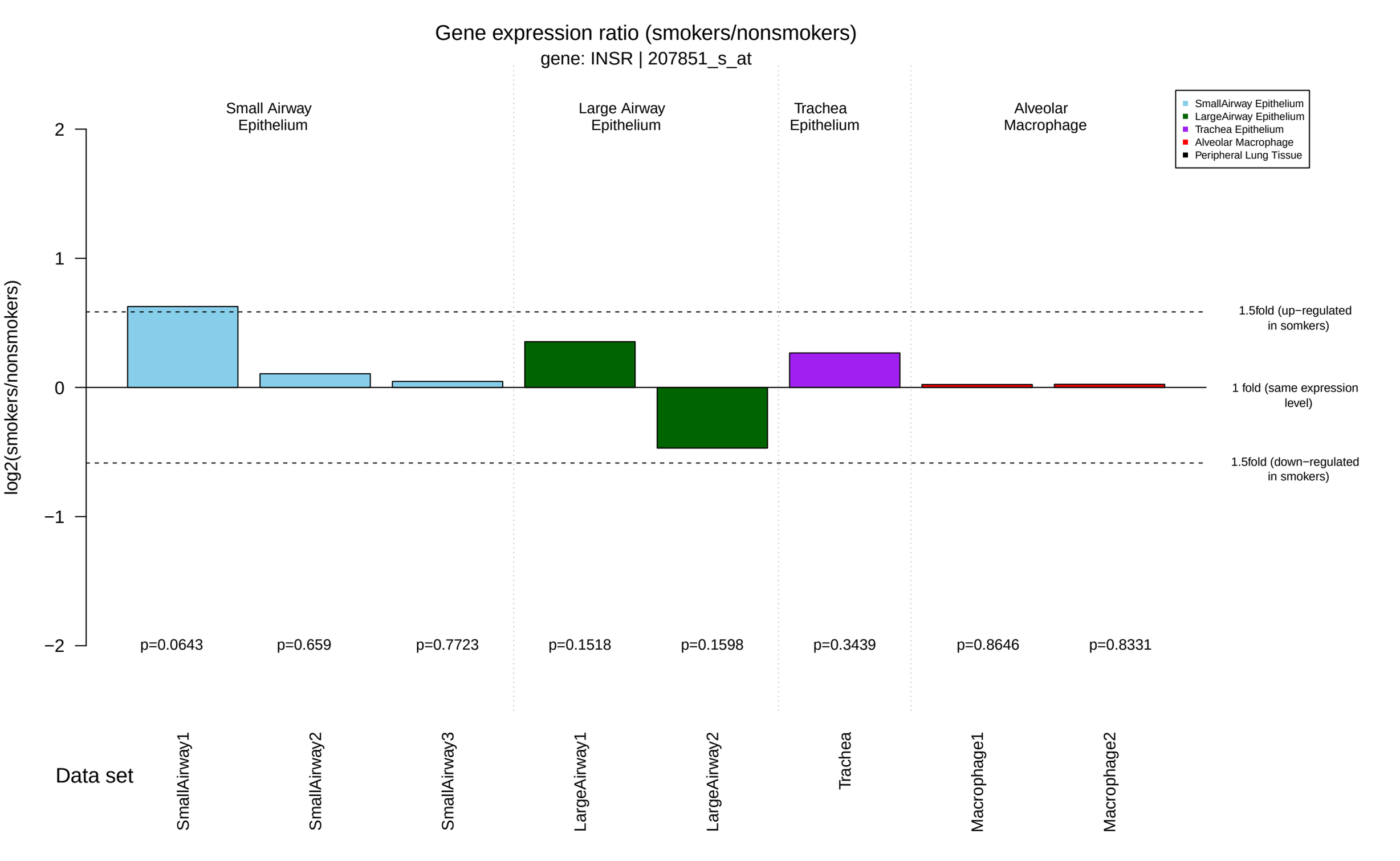
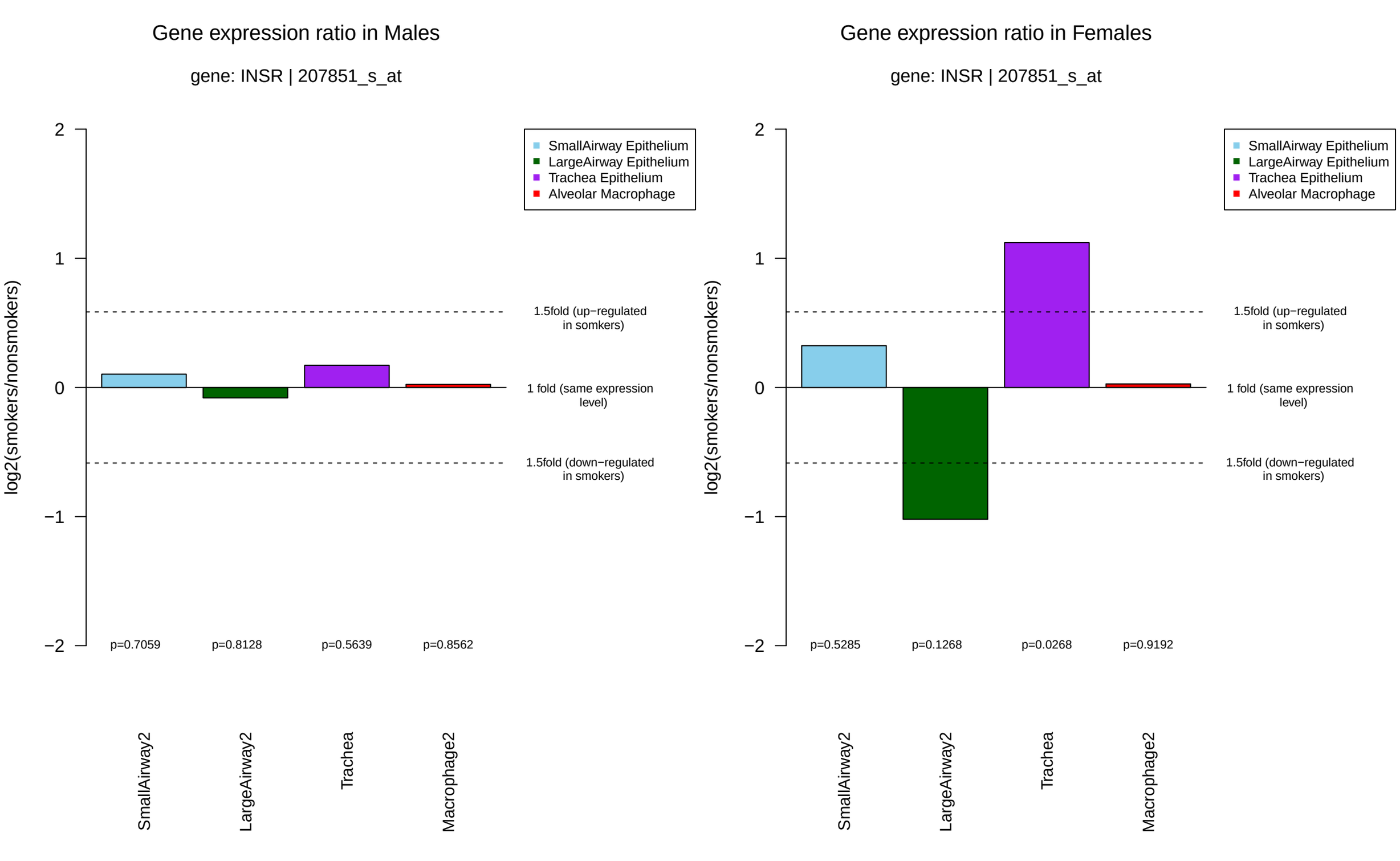
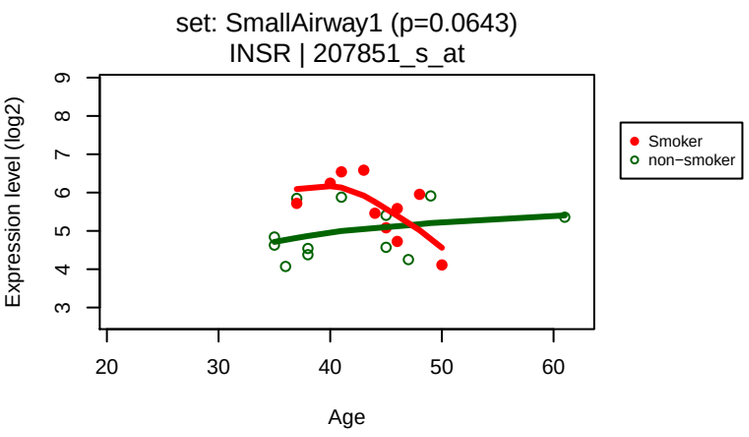
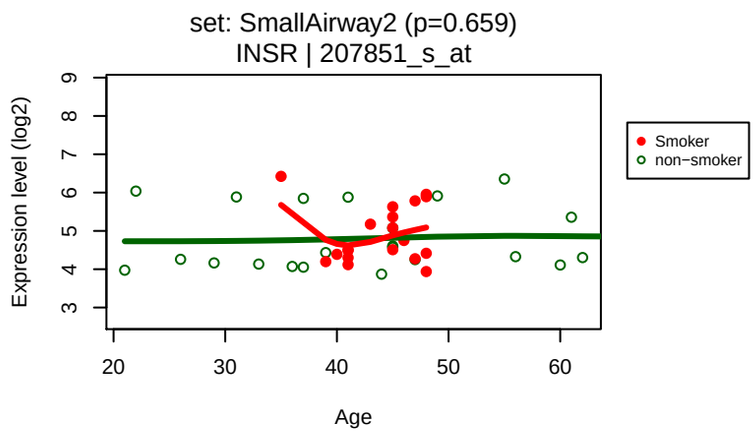
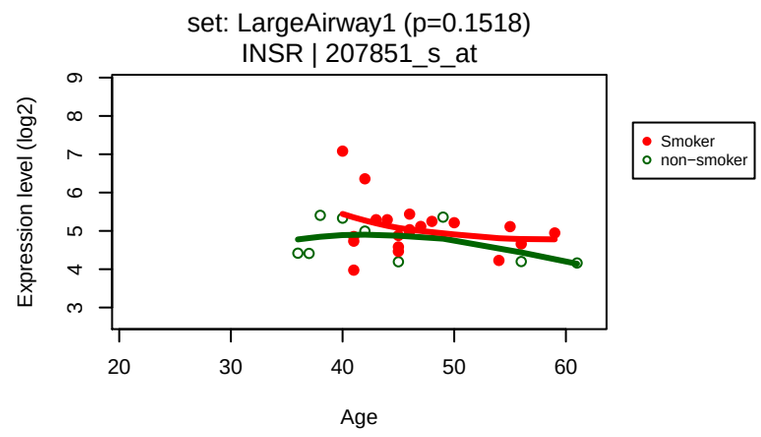

Gene expression of smokers and nonsmokers
gene: INSR | 207851_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0643
p=0.659
p=0.7723
p=0.1518
p=0.1598
p=0.3439
p=0.8646
p=0.8331
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

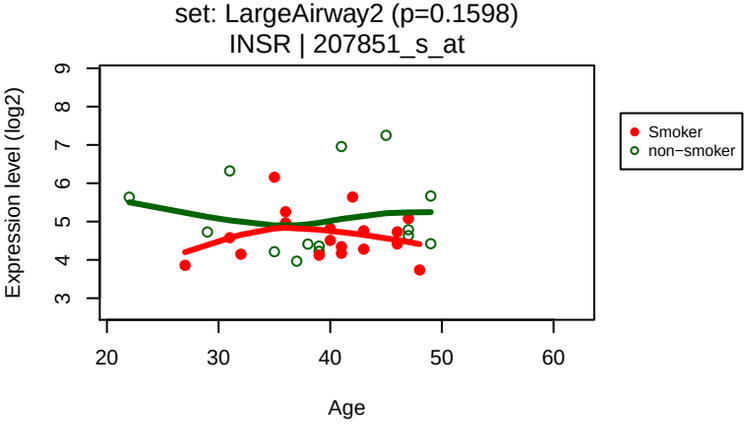

Gene expression ratio (smokers/nonsmokers)
gene: INSR | 207851_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0643
p=0.659
p=0.7723
p=0.1518
p=0.1598
p=0.3439
p=0.8646
p=0.8331
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

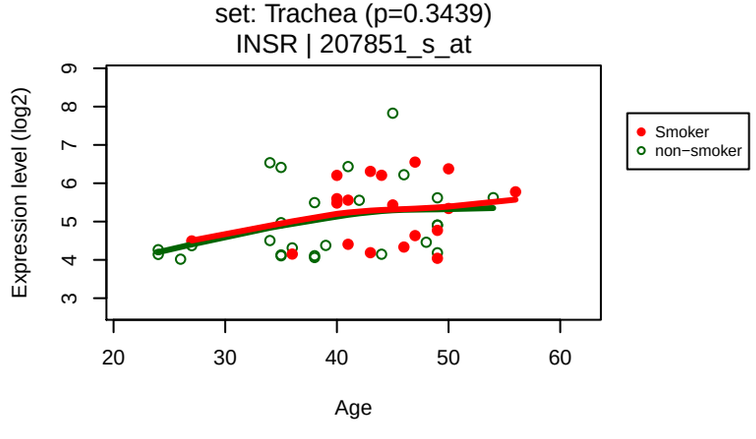

Gene expression ratio in Males
gene: INSR | 207851_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7059
p=0.8128
p=0.5639
p=0.8562
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: INSR | 207851_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5285
p=0.1268
p=0.0268
p=0.9192
SmallAirway2
LargeAirway2
Trachea
Macrophage2

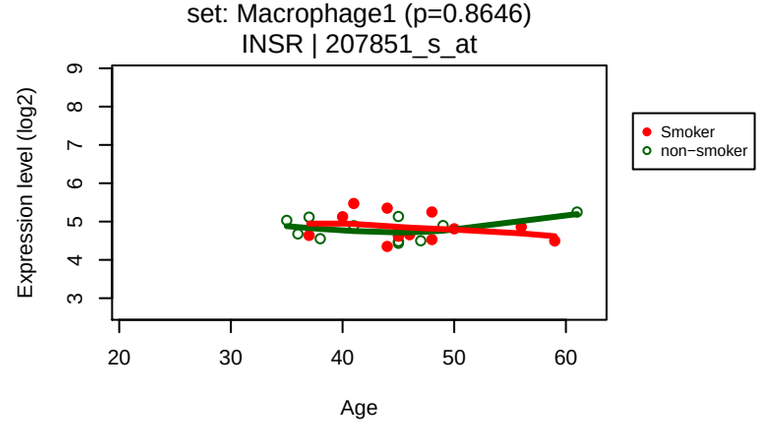

set: SmallAirway1 (p=0.0643)
INSR | 207851_s_at
Expression level (log2)
Age
Smoker
non−smoker

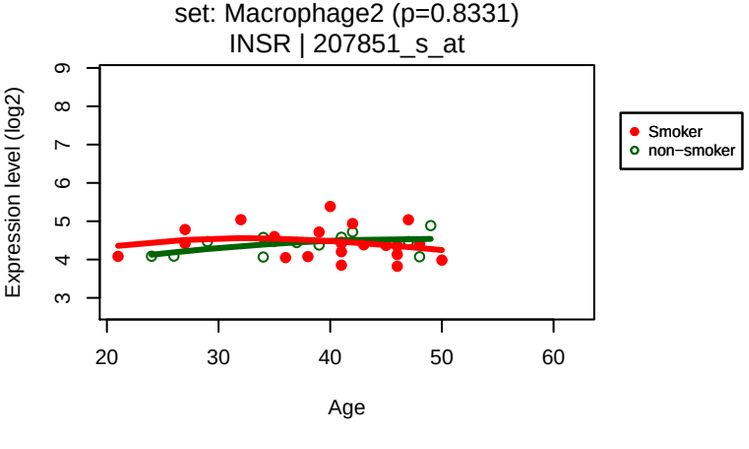

set: SmallAirway2 (p=0.659)
INSR | 207851_s_at
Expression level (log2)
Age
Smoker
non−smoker

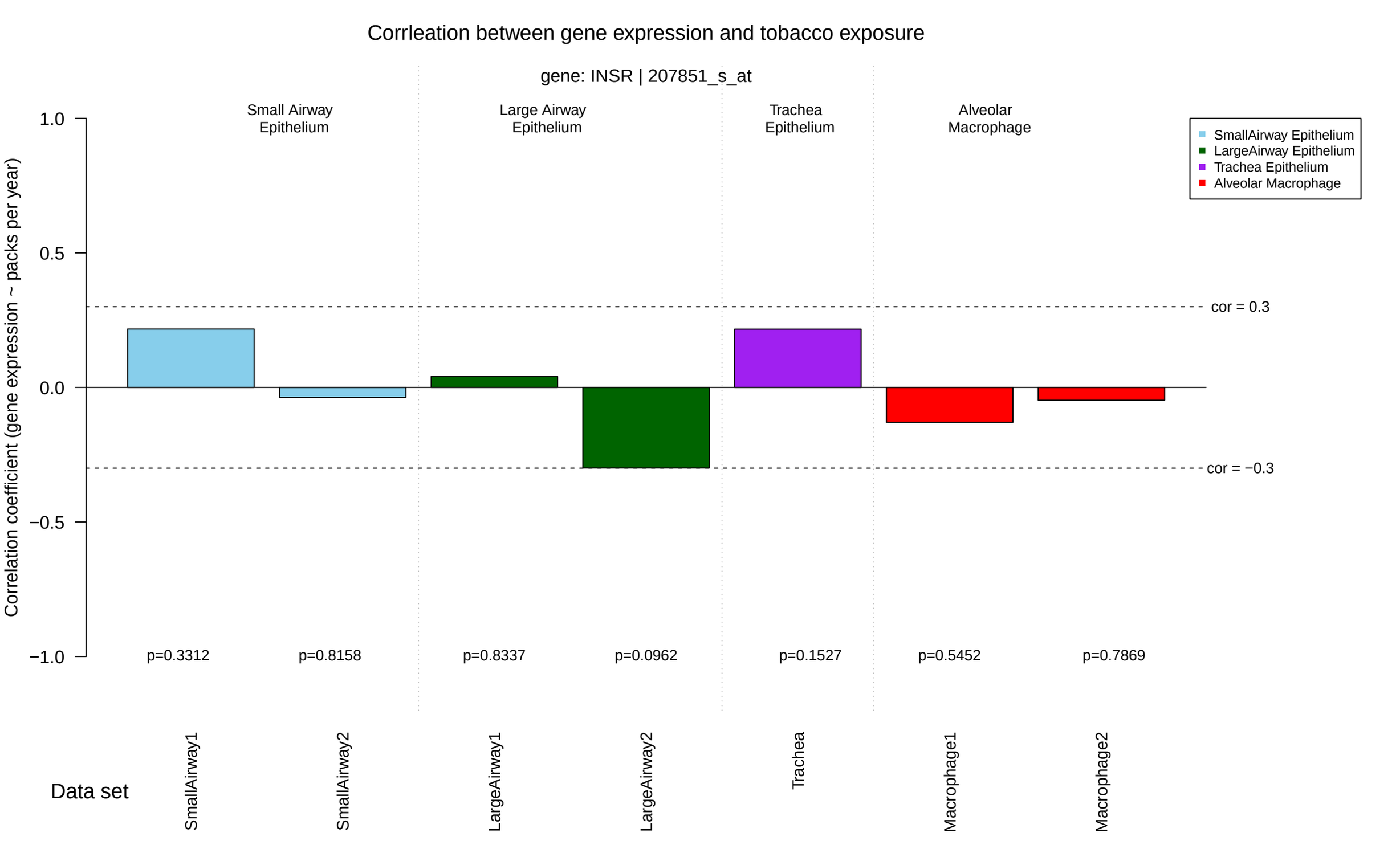

set: LargeAirway1 (p=0.1518)
INSR | 207851_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1598)
INSR | 207851_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3439)
INSR | 207851_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8646)
INSR | 207851_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8331)
INSR | 207851_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: INSR | 207851_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3312
p=0.8158
p=0.8337
p=0.0962
p=0.1527
p=0.5452
p=0.7869
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2